AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XII

ANNUAL EXAMINATIONS (THEORY) 2025

Mathematics Paper II

Time: 1 hour 30 minutes Marks: 50

AKU-EB Annual Examinations 2025 for Teaching & Learning only

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

I agree that this is my name and school. **Candidate's Signature**

RUBRIC

2. There are NINE questions. Answer ALL questions. Choices are specified inside the paper.

3. When answering the questions:

   Read each question carefully.
   Use black pointer to write your answers. DO NOT write your answers in pencil.
   Use a black pencil for diagrams. DO NOT use coloured pencils.
   DO NOT use staples, paper clips, glue, correcting fluid or ink erasers.
   Complete your answer in the allocated space only. DO NOT write outside the answer box.

4. The marks for the questions are shown in brackets ( ).

5. A formulae list is provided on page 2 and 3. You may refer to it during the paper, if you wish.

6. You may use a scientific calculator if you wish.

List of Formulae

Note:

- The symbols used in the formulae have their usual meaning.

Functions and Limits

$$\lim_{\theta\to 0}\frac{\sin\theta}{\theta}=1 \qquad \lim_{x\to 0}\frac{\sqrt{x+a}-\sqrt{a}}{x}=\frac{1}{2\sqrt{a}} \qquad \lim_{x\to a}\frac{x^n-a^n}{x-a}=na^{n-1}$$

$$\lim_{x\to 0}(1+x)^{\frac{1}{x}}=e \qquad \lim_{x\to+\infty}\left(1+\frac{1}{x}\right)^x=e \qquad \lim_{x\to 0}\frac{a^x-1}{x}=\log_e a$$

Differentiation

$$\frac{\Delta y}{\Delta x}=\frac{f(x+h)-f(x)}{h} \qquad \frac{d}{dx}(\sec x)=\sec x\tan x \qquad \frac{d}{dx}x^n=nx^{n-1} \qquad \frac{d}{dx}(\sin x)=\cos x$$

$$\frac{d}{dx}(\cos x)=-\sin x \qquad \frac{d}{dx}(\tan x)=\sec^2 x \qquad \frac{d}{dx}(\cot x)=-\operatorname{cosec}^2 x \qquad \frac{d}{dx}(\operatorname{cosec} x)=-\operatorname{cosec} x\cot x$$

$$\frac{d}{dx}\left[\sin^{-1}x\right]=\frac{1}{\sqrt{1-x^2}},\ x\in(-1,1) \qquad \frac{d}{dx}\left[\cos^{-1}x\right]=-\frac{1}{\sqrt{1-x^2}},\ x\in(-1,1) \qquad \frac{d}{dx}\left[\tan^{-1}x\right]=\frac{1}{x^2+1},\ x\in R \qquad \frac{d}{dx}\left[\sec^{-1}x\right]=\frac{1}{|x|\sqrt{x^2-1}},\ x\in[-1,1]'$$

$$\frac{d}{dx}\left[\operatorname{cosec}^{-1}x\right]=-\frac{1}{|x|\sqrt{x^2-1}},\ x\in[-1,1]' \qquad \frac{d}{dx}\left[\cot^{-1}x\right]=-\frac{1}{1+x^2},\ x\in R \qquad \frac{d}{dx}(a^x)=a^x\ln a \qquad \frac{d}{dx}(\log_a x)=\frac{1}{x}\times\frac{1}{\ln a}$$

$$\frac{d}{dx}(\sinh x)=\cosh x \qquad \frac{d}{dx}(\cosh x)=\sinh x \qquad \frac{d}{dx}(\tanh x)=\operatorname{sech}^2 x$$

$$\frac{d}{dx}(\operatorname{cosech} x)=-\coth x\operatorname{cosech} x \qquad \frac{d}{dx}(\operatorname{sech} x)=-\tanh x\operatorname{sech} x \qquad \frac{d}{dx}(\coth x)=-\operatorname{cosech}^2 x$$

$$\frac{d}{dx}\left[f(x)\times g(x)\right]=f(x)\frac{d}{dx}\left[g(x)\right]+g(x)\frac{d}{dx}\left[f(x)\right] \qquad \frac{d}{dx}\left(\frac{f(x)}{g(x)}\right)=\frac{g(x)\frac{d}{dx}\left[f(x)\right]-f(x)\frac{d}{dx}\left[g(x)\right]}{\left[g(x)\right]^2}$$

Maclaurin Series $f(x)=f(0)+f'(0)x+\frac{f''(0)}{2!}x^2+\frac{f'''(0)}{3!}x^3+\ldots+\frac{f^n(0)}{n!}x^n+\ldots$

Taylor's Series $f(x)=f(a)+f'(a)(x-a)+\frac{f''(a)}{2!}(x-a)^2+\frac{f'''(a)}{3!}(x-a)^3+\frac{f^{iv}(a)}{4!}(x-a)^4+\ldots+\frac{f^n(a)}{n!}(x-a)^n+\ldots$

Integration

$$\int f'(x)dx=f(x)+c \qquad \int[f(x)]^n f'dx=\frac{[f(x)]^{n+1}}{n+1}+c,\quad (n\neq -1) \qquad \int f(x)g'(x)dx=f(x)g(x)-\int f'(x)g(x)dx$$

$$\int\frac{f'(x)}{f(x)}dx=\ln f(x)+c \qquad \int\operatorname{cosec} xdx=\ln|\operatorname{cosec} x-\cot x|+c \qquad \int\sec xdx=\ln|\sec x+\tan x|+c$$

Plane Analytical Geometry (Straight Line)

$$d=\frac{|ax_1+by_1+c|}{\sqrt{a^2+b^2}} \qquad y-y_1=\frac{y_2-y_1}{x_2-x_1}(x-x_1) \qquad \left(\frac{x_1+x_2}{2},\frac{y_1+y_2}{2}\right)$$

Annual Examinations 2025 AKU-EB for Teaching & Learning only

Point of internal division

$$\left(\frac{k_1x_2+k_2x_1}{k_1+k_2},\ \frac{k_1y_2+k_2y_1}{k_1+k_2}\right)$$

Point of external division

$$\left(\frac{k_1x_2-k_2x_1}{k_1-k_2},\ \frac{k_1y_2-k_2y_1}{k_1-k_2}\right)$$

$$\frac{x-x_1}{\cos\alpha}=\frac{y-y_1}{\sin\alpha}=r \text{ (say)}$$

$$d=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2}$$

$$\frac{x}{a}+\frac{y}{b}=1$$

$$x\cos\alpha+y\sin\alpha=p$$

$$y=mx+c$$

$$y-y_1=m(x-x_1)$$

$$\theta=\tan^{-1}\left[\frac{m_2-m_1}{1+m_1m_2}\right]$$

$$\theta=\tan^{-1}\left[\frac{2\sqrt{h^2-ab}}{a+b}\right]$$

$$\begin{vmatrix}a_1 & b_1 & c_1\\ a_2 & b_2 & c_2\\ a_3 & b_3 & c_3\end{vmatrix}=0$$

$$\Delta=\frac{1}{2}\begin{vmatrix}x_1 & y_1 & 1\\ x_2 & y_2 & 1\\ x_3 & y_3 & 1\end{vmatrix}$$

Circles

$$(x-h)^2+(y-k)^2=r^2$$

$$x^2+y^2+2gx+2fy+c=0$$

Equation of normal to a circle

$$(y-y_1)(x_1+g)=(x-x_1)(y_1+f)$$

Equation of tangent to a circle

$$xx_1+yy_1+g(x+x_1)+f(y+y_1)+c=0$$

Length of tangent to a circle from a point (x_1, y_1), $l=\sqrt{x_1^2+y_1^2+2gx_1+2fy_1+c}$

Parabola

$$x^2=4ay \qquad (x-h)^2=4a(y-k) \qquad y^2=4ax \qquad (y-k)^2=4a(x-h)$$

Ellipse

$$\frac{(x-h)^2}{a^2}+\frac{(y-k)^2}{b^2}=1,\ a>b \qquad b^2=a^2(1-e^2) \qquad c=ae \qquad \frac{(y-k)^2}{a^2}+\frac{(x-h)^2}{b^2}=1,\ a>b$$

Hyperbola

$$\frac{(x-h)^2}{a^2}-\frac{(y-k)^2}{b^2}=1 \qquad b^2=a^2(e^2-1) \qquad c=ae \qquad \frac{(y-k)^2}{a^2}-\frac{(x-h)^2}{b^2}=1$$

Translation and Rotation

$$x=X+h,\ y=Y+k \qquad X=x\cos\theta+y\sin\theta,\ Y=y\cos\theta-x\sin\theta \qquad \tan 2\theta=\frac{2h}{a-b}$$

Vectors

$$\underline{u}.\underline{v}=|\underline{u}||\underline{v}|\cos\theta$$

Area of a triangle $=\frac{1}{2}|\underline{u}\times\underline{v}|$

$$\underline{u}\times\underline{v}=\left(|\underline{u}||\underline{v}|\sin\theta\right)\hat{\underline{n}} \qquad \underline{r}=\frac{q\underline{a}+p\underline{b}}{p+q}$$

$$\left|\overrightarrow{P_1P_2}\right|=\sqrt{(x_2-x_1)^2+(y_2-y_1)^2+(z_2-z_1)^2}$$

$$\frac{1}{6}(\underline{u}\times\underline{v}).\underline{w}=\frac{1}{6}[\underline{u}\ \underline{v}\ \underline{w}] \qquad \underline{u}.(\underline{v}\times\underline{w})=\begin{vmatrix}a_1 & b_1 & c_1\\ a_2 & b_2 & c_2\\ a_3 & b_3 & c_3\end{vmatrix}$$

Some Trigonometric Identities

$$\cos(\alpha\pm\beta)=\cos\alpha\cos\beta\mp\sin\alpha\sin\beta$$

$$\sin(\alpha\pm\beta)=\sin\alpha\cos\beta\pm\cos\alpha\sin\beta$$

$$\cos P+\cos Q=2\cos\frac{P+Q}{2}\cos\frac{P-Q}{2}$$

$$\cos P-\cos Q=-2\sin\frac{P+Q}{2}\sin\frac{P-Q}{2}$$

$$\sin P+\sin Q=2\sin\frac{P+Q}{2}\cos\frac{P-Q}{2}$$

$$\sin P-\sin Q=2\cos\frac{P+Q}{2}\sin\frac{P-Q}{2}$$

$$\sin 2\alpha=2\sin\alpha\cos\alpha$$

$$\cos 2\alpha=\cos^2\alpha-\sin^2\alpha$$

Annual Examinations 2025 AKU-EB for Teaching & Learning only

Q.1. (Total 3 Marks)

Evaluate the limit of the given function.

$$\lim_{x \to +\infty} \left(1 + \frac{1}{2x}\right)^{-6x}$$

AKU-EB Annual Examinations 2025 for Teaching & Learning only

(ATTEMPT ANY TWO PARTS a, b OR c OF Q.2.)

Q.2. (Total 10 Marks)

a. Using differentiation by first principle, differentiate $y = \sin(1-2x)$. (5 Marks)

AKU-EB Annual Examinations 2025 for Teaching & Learning only

(ATTEMPT ANY TWO PARTS a, b OR c OF Q.2.)

b.

i. If $y = \dfrac{e^{ax} - e^{-ax}}{1 + \ln ax}$, then find $\dfrac{dy}{dx}$. (3 Marks)

ii. If $f(x) = \sinh^{-1}\left(\dfrac{a}{x}\right)$, then find $f'(x)$. (2 Marks)

(**Note:** $\dfrac{d}{dx}\sinh^{-1} x = \dfrac{1}{\sqrt{1+x^2}}$)

AKU-EB Annual Examinations 2025 for Teaching & Learning only

(ATTEMPT ANY TWO PARTS a, b OR c OF Q.2.)

c. A function is defined as $y = \frac{1}{3}x^3 - x^2 - 3x + 5$. Find the value(s) of x at which the

i. tangent is parallel to x-axis. (3 Marks)

ii. function obtains the relative highest value. (2 Marks)

AKU-EB Annual Examinations 2025 for Teaching & Learning only

(ATTEMPT ANY TWO PARTS a, b OR c OF Q.3.)

Q.3. (Total 10 Marks)

a. Using integration by substitution, evaluate $\int \frac{dx}{\sqrt{x^2+4x+3}}$. (5 Marks)

AKU-EB Annual Examinations 2025 for Teaching & Learning only

(ATTEMPT ANY TWO PARTS a, b OR c OF Q.3.)

b. Apply the method of partial fraction to evaluate the given integral. (5 Marks)

$$\int \frac{x^2}{(x-4)(x+5)^2}\,dx$$

AKU-EB Annual Examinations 2025 for Teaching & Learning only

(ATTEMPT ANY TWO PARTS a, b OR c OF Q.3.)

c.

i. Find the particular solution of the given differential equation. (4 Marks)

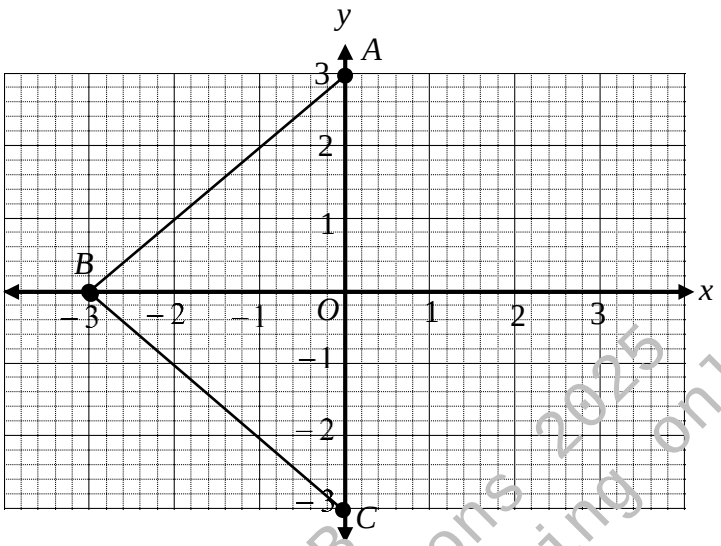
$$\left(x^2+1\right)\frac{dy}{dx}=1; \qquad y(0)=3$$

AKU-EB Annual Examinations 2025 for Teaching & Learning only

ii. Hence, find the value of x for $y = 0.5$ (1 Mark)

Q.4. (Total 4 Marks)

In the given diagram, *ABC* is a triangle with the given vertices on the Cartesian coordinate system.



AKU-EB Annual Examinations 2025 for Teaching & Learning only

Using the given information, find the

a. equation of the side *AB*. (2 Marks)

b. equation of the altitude from C to *AB*. (2 Marks)

Q.5. (Total 4 Marks)

Using the given graph, find the maximum value of the objective function by shading the feasible region in the given linear programing problem.

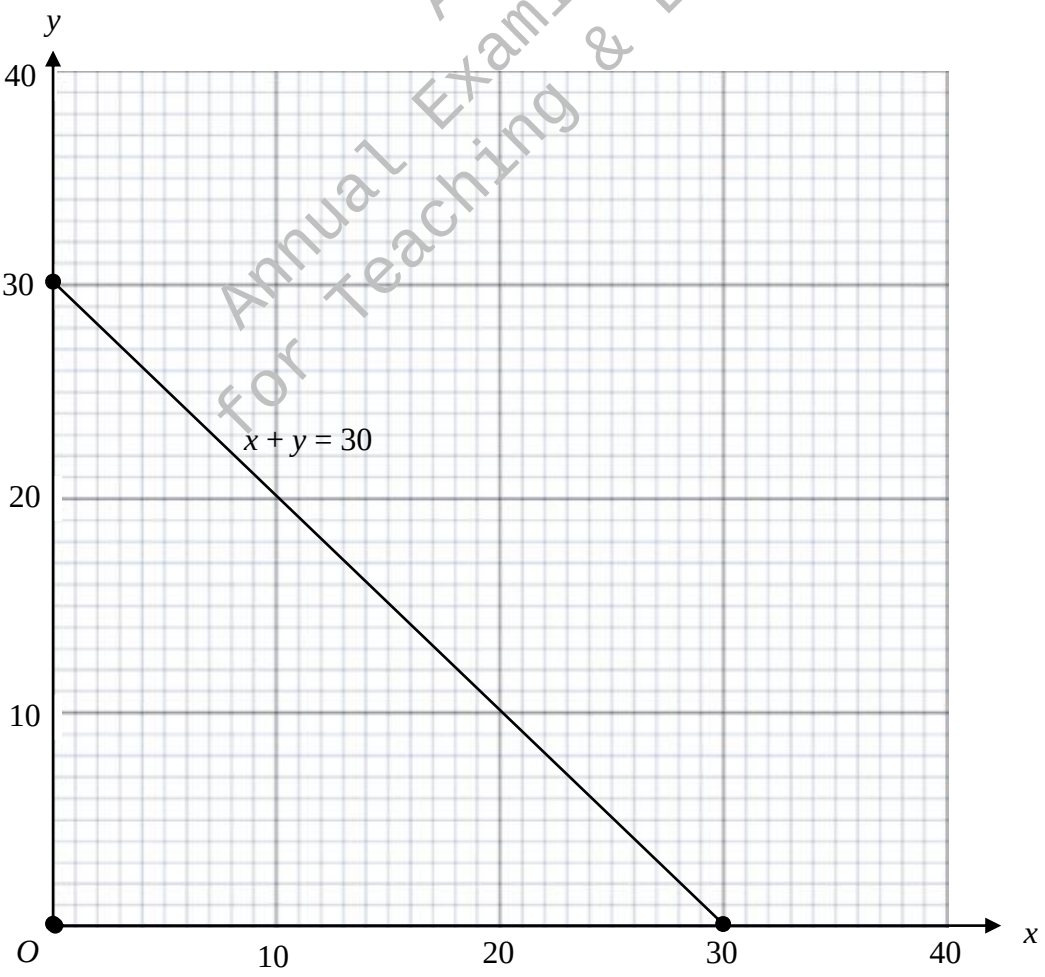
$P(x, y) = 300x + 200y$

$x + y \leq 30$

$x + 2y \leq 40$

$x \geq 0,\ y \geq 0$

AKU-EB Annual Examinations 2025 for Teaching & Learning only



Q.6. (Total 4 Marks)

The general equation of a circle is given as $x^2 + y^2 + 2gx + my = 7$. If the centre of the circle is (p, 9) and radius is 13 units, then find the value of

a. *m.* (2 Marks)

b. *p*, where *p* is positive. (2 Marks)

AKU-EB Annual Examinations 2025 For Teaching & Learning only

(ATTEMPT ANY TWO PARTS a, b OR c OF Q.7.)

Q.7. (Total 8 Marks)

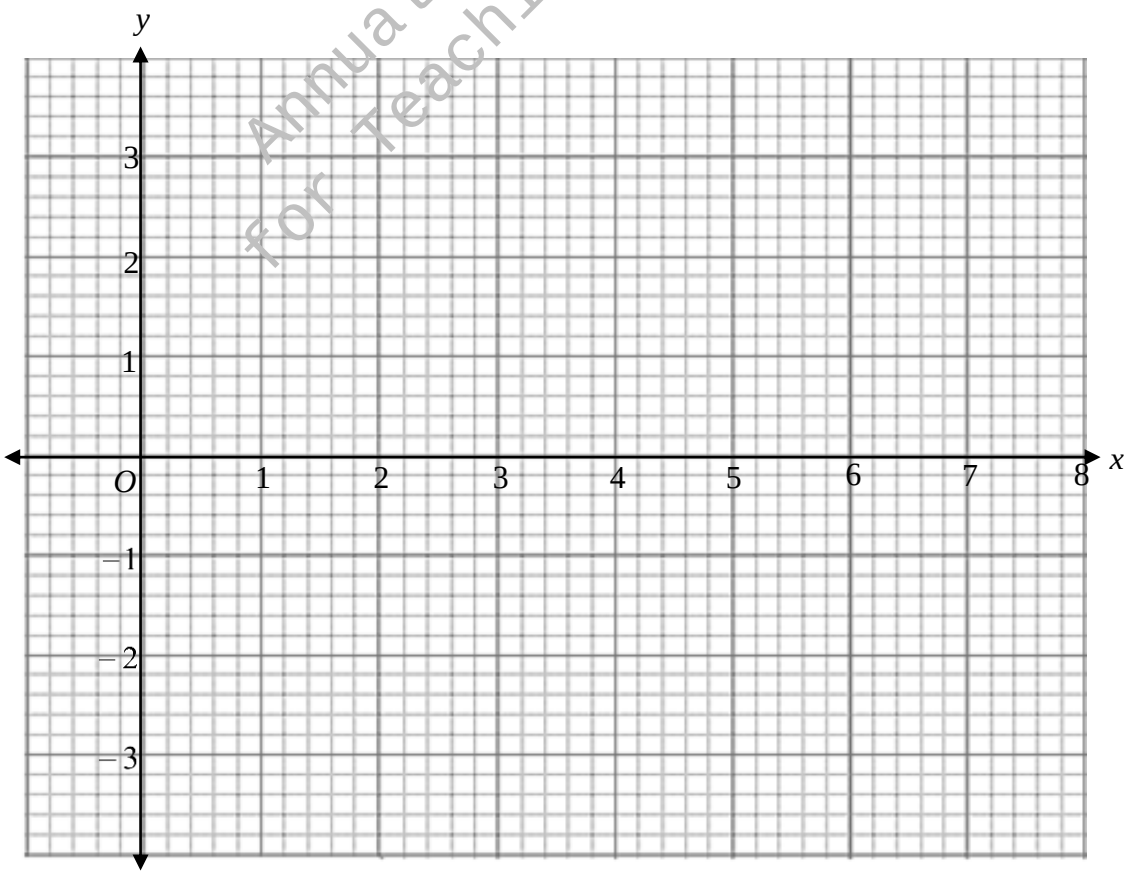
a. A parabola is represented by the equation $y^2 = 12(x-2)$.

i. Find the following elements of the parabola.

I. Vertex (1 Mark)

II. Focus (1 Mark)

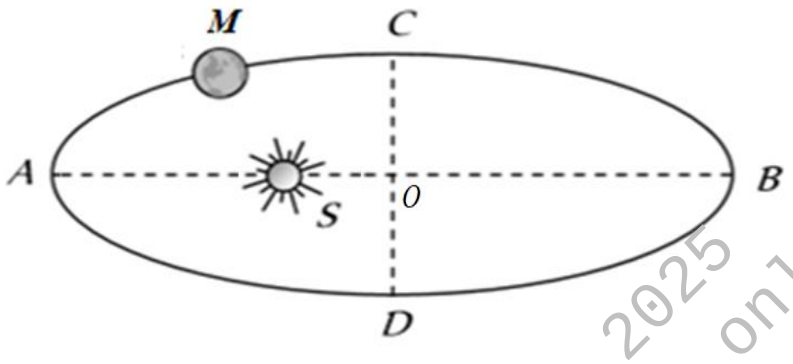
ii. Sketch the parabola and label the elements found in part i. (2 Marks)



AKU-EB Annual Examinations 2025 for Teaching & Learning only

(ATTEMPT ANY TWO PARTS a, b OR c OF Q.7.)

b. The planet Mars revolves around the Sun in an elliptic orbit. During its revolution, Mars keeps the minimum distance of 129.5 million kilometres and maximum distance of 154.4 million kilometres from the Sun. The situation is depicted in the given diagram.



Assuming the Sun as on one of the foci, find the

i. location of the Sun. (2 Marks)

ii. equation representing the orbit of Mars. (2 Marks)

AKU-EB Annual Examinations 2025 for Teaching & Learning only

(ATTEMPT ANY TWO PARTS a, b OR c OF Q.7.)

c. Find the equations of tangent and normal to the hyperbola $x^2 - 4y^2 + 10x + 24y + 5 = 0$ at point (– 5, 1). (4 Marks)

AKU-EB Annual Examinations 2025 for Teaching & Learning only

Q.8. (Total 4 Marks)

Find the transformed equation of $4x^2 - 4y = 0$ with respect to the new axes obtained by rotation of axes about the origin through an angle of $60°$.

AKU-EB Annual Examinations 2025 for Teaching & Learning only

(ATTEMPT EITHER PART a OR PART b OF Q.9.)

Q.9. (Total 3 Marks)

a. Find the value of x for which the two vectors $\vec{A} = x\hat{i} - 5\hat{j}$ and $\vec{B} = \hat{i} - 3\hat{j} + 5\hat{k}$ are perpendicular to each other.

b. Find a vector perpendicular to the plane containing points $A(-1, 3, 0)$, $B(1, -1, 2)$ and $C(1, 3, -1)$.

AKU-EB Annual Examinations 2025 for Teaching & Learning only

END OF PAPER

Please use this page for rough work

AKU-EB
Annual Examinations 2025
for Teaching & Learning only

Please use this page for rough work

AKU-EB
Annual Examinations 2025
for Teaching & Learning only